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Candidate

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Session

ID: U8GMNS-S9G
Time limit: 40 min.
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Notes:

N/A

Similarity Check

Status: not found
No similar solutions have been detected.

Test score

0%

Tasks in test

1 | CustomCollection

Score

N/A

Tasks Details

Medium

1. CustomCollection

Implement a custom collection with a simplified interface.

Task Score	Correctness	Performance
	Not assessed	Not assessed

Task description

Your goal is to implement a custom collection with a simplified interface. Your class should be called `StringMap` and should allow users to store a collection of pairs of the form `<key, value>`. The key will always be a string and the value can be any reference type. `StringMap` does not have to be thread-safe; however, it must implement the following interface:

```
public interface IStringMap<TValue> where TValue: class
{
    /// <summary>Returns number of elements in a map</summary>
    int Count { get; }

    /// <summary>
    /// If <c>GetValue</c> method is called but a given key is not in a map then <c>DefaultValue</c> is returned
    /// </summary>
    TValue DefaultValue { get; set; }

    /// <summary>
    /// Adds a given key and value to a map.
    /// If the given key already exists in the map, then the value associated with this key should be overridden
    /// </summary>
    /// <returns>true if the value for the key was overridden, otherwise false</returns>
    /// <exception cref="System.ArgumentNullException">If the key is null</exception>
    /// <exception cref="System.ArgumentException">If the key is an empty string</exception>
    /// <exception cref="System.ArgumentNullException">If the value is null</exception>
    bool AddElement(string key, TValue value);

    /// <summary>
    /// Removes a given key and associated value from a map.
    /// </summary>
    /// <returns>true if the key was in the map and was removed, otherwise false</returns>
    /// <exception cref="System.ArgumentNullException">If the key is null</exception>
    /// <exception cref="System.ArgumentException">If the key is an empty string</exception>
    bool RemoveElement(string key);

    /// <summary>
    /// Returns the value associated with a given key.
    /// </summary>
    /// <returns>The value associated with a given key or <c>DefaultValue</c> if the key does not exist in a map
    /// <exception cref="System.ArgumentNullException">If the key is null</exception>
    /// <exception cref="System.ArgumentException">If the key is an empty string</exception>
    TValue GetValue(string key);
}
```

Solution

[See Live Version](#)

Programming language used: (None)

No final submissions.